

Question

The carbonyl compound **A** of a widely distributed common metallic element **M** can react with N_2O_4 to obtain a compound **B** of the uncommon +1 valence **M** and release a colorless gas. **A** can also react directly with bromine to obtain another +1 valence compound **C** of **M**. Electrolysis of an aqueous solution of the common valence acetate of **M** can yield a multinuclear complex whose cation **D** contains a 3-coordinate oxygen atom and the mass fraction of **M** is 30.80%.

Which of the following options is correct?

- A.** **M** is located in Group IB
- B.** The molecular formula of **A** can be expressed as $\text{M}(\text{CO})_6$
- C.** **A** contains bridging carbonyl ligands.
- D.** **A** can react with N_2O_4 to produce **B** while releasing CO
- E.** In the periodic table, the metal **M'** located directly below and adjacent to **M** has stable non-radioactive isotopes.
- F.** The reaction of the highest-valent oxyanion of **M** with sulfite yields an unstable oxyanion containing **M**. The sodium salt of this oxyanion can disproportionate in aqueous solution. Balance the disproportionation equation and simplify the coefficients to the simplest integer ratio, where the sum of the coefficients on the left side of the equation is greater than the sum of the coefficients on the right side.
- G.** The valence of cation **D** is +1.
- H.** All of the above options are incorrect.

Answer

Correct Answer: **G**

Detailed Explanation

The question mentions that **D** is a polynuclear coordination complex cation, containing a 3-coordinate oxygen atom ($\mu_3 - O$), and the mass fraction of **M** in **D** is 30.80%. Polynuclear metal acetate complexes with $\mu_3 - O$ bridging oxygen atoms are a classic structure, and their general formula is usually $[M_3(\mu_3 - O)(\mu_2 - CH_3COO)_6]^{n+}$. The molar mass of cation **D** $= 3 \times Ar(\mathbf{M}) + 370.26$, thus we get the equation:

$$3 \times Ar(\mathbf{M}) = 0.308 \times (3 \times Ar(\mathbf{M}) + 370.26)$$

Solving for the mass of **M** gives 54.93, and the metal element corresponding to this mass is **Mn**, so A is incorrect.

CHECKPOINT

1 PTS

M is Mn

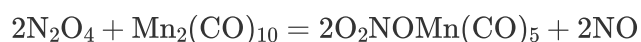
The molecular formula of the carbonyl compound **A** of the metal element **M** should be $Mn_2(CO)_{10}$, which is axially connected by two $(CO)_5Mn$ subunits through a $Mn - Mn$ metal bond, without bridging carbonyl ligands, so B and C are incorrect.

CHECKPOINT

1 PTS

A is $\text{Mn}_2(\text{CO})_{10}$

A can react with N_2O_4 to obtain a compound **B** of the uncommon +1 valence **M** and release a colorless gas. The metal valence in $\text{Mn}_2(\text{CO})_{10}$ is 0, and it is oxidized in the reaction. The equation is



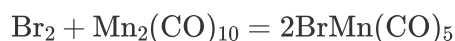
B is $\text{O}_2\text{NOMn}(\text{CO})_5$. In the process of **A** reacting with N_2O_4 to generate **B**, NO is released, so D is incorrect.

CHECKPOINT

1 PTS

B is $\text{O}_2\text{NOMn}(\text{CO})_5$

A can also directly react with bromine to obtain another +1 valence compound **C** of **M**. The equation is



C is $\text{BrMn}(\text{CO})_5$

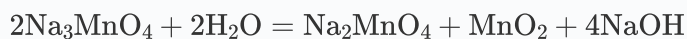
The metal **M'** located directly below **M** and adjacent to it in the periodic table is Tc, which is the first metal prepared by artificial methods. It has no stable non-radioactive isotopes, only radioactive isotopes, so E is incorrect.

CHECKPOINT

1 PTS

M' is Tc, has no stable non-radioactive isotopes

Reacting the highest-valent oxyacid radical of **M** with sulfite can obtain an unstable oxyacid radical containing **M**. In the highest-valent oxyacid radical, Mn has a +7 valence, and after reacting with sulfite, it becomes +5 valence. The sodium salt of this oxyacid radical can disproportionate in aqueous solution. The equation is

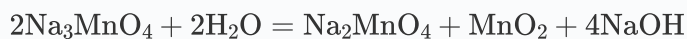


, the sum of coefficients on the left side is less than the sum of coefficients on the right side, so F is incorrect.

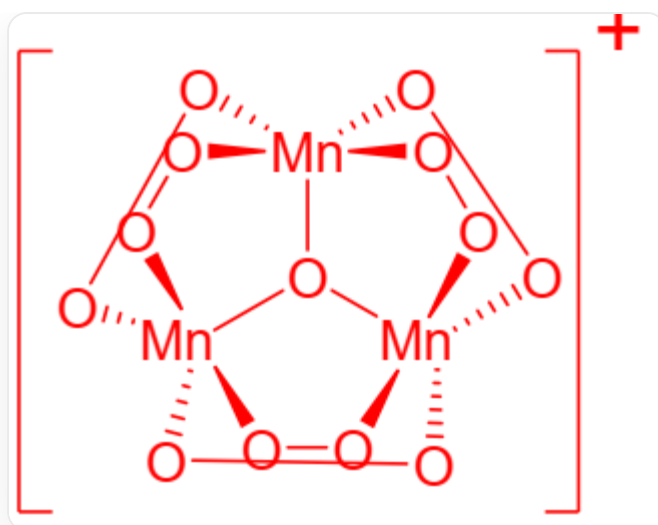
CHECKPOINT

1 PTS

Disproportionation equation is



The structure of cation **D** is shown in the figure



Structure of cation **D**, the structural formula is $[\text{Mn}_3(\mu_3 - \text{O})(\mu_2 - \text{CH}_3\text{COO})_6]^+$, $\text{O} - \text{O}$ represents the acetate bidentate ligand, which contains 3 Mn, 1 $\mu_3 - \text{O}$ and 6 acetate ligands. Each Mn is connected to 4 O of acetate and $\mu_3 - \text{O}$, and the valence is +1

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CHECKPOINT

1 PTS

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